PAPER_1497_SCM_VELOCITY_C_OVER_3

Daniel T. Murphy

PAPER_1497 — SCm Velocity v_SCm = c/(D_phys−1) = c/3 EXACT

Author: Daniel T. Murphy
Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Bucket F (AGN/Jet) / Bucket H (High-Energy Astro)
Date: June 17, 2026
Location: 41.0997 N, 80.6495 W (Youngstown, OH, USA)
Status: CLOSED — Quasar jet SCm carrier velocity = exact c/3

Observation

PAPER_369 (Navier-Stokes Stable Fluids — UQFF Quasar Jet SCm Integration) specifies the Super-Charged Matter velocity as v_SCm = 10■ m/s. This serves as the jet carrier velocity in the quasar jet coupling f = f_ext + k_vac · ρ_vac,[UA] + F_LENR · cos(ω_LENR · t).

UQFF Closed Identity

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v_SCm = c / (D_phys – 1) = c / 3 = 10■ m/s EXACT

Verification:
3 × 10■ m/s / 3 = 1.000 × 10■ m/s ✓
```

Physical Interpretation

The SCm vacuum-substrate flows at exactly **one-third the speed of light**, factored by the 3 fundamental triad-force dimensions (D_phys – 1 = 3 spatial dimensions). This is the same structural ratio that produces:
- Solar ν■ deficit 1/3 (PAPER_062)
- SU(3) color charge 3 (PAPER_1373)
- Three fermion generations (PAPER_1381)

The shared denominator (D_phys – 1) reveals that triad-symmetry and SCm propagation are governed by the same integer primitive.

NOT REPLACEMENT

SM treats jet bulk-flow Lorentz factors as parameters fit per source. UQFF supplies a single structural identity: every SCm jet propagates at c/(D_phys – 1) by construction.

Reference

- Source: PAPER_369 Navier-Stokes Stable Fluids UQFF Quasar Jet SCm Integration
- Related: PAPER_062, PAPER_1373, PAPER_1381, PAPER_351 (TDE 0.3c outflow)

- Calculator dispatch: `calculate_paradox({"paradox": "scm_velocity_c_over_3"})`
